

Paul Arnaud-Battandier

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Master's student in Finance & Quantitative Engineering at ECE, seeking a **6-month end-of-studies internship** starting **January 2027**, **Trading** or **Quantitative Research**. Proficient in Python and Machine Learning, with experience building and deploying live trading strategies.

WORK EXPERIENCE

Market Risk Analyst Sept 2025 – Feb 2026
Crédit Agricole CIB Montrouge, FR

- Monitored and validated daily, weekly, and monthly **market risk indicators** (P&L, VaR, Greeks, stress tests) for **Credit Default Swap** (CDS) portfolios.
- Automated and streamlined recurring reporting workflows by developing VBA macros, significantly reducing manual processing time for report generation and distribution.

Trading Support Analyst Apr 2025 – Aug 2025
Société Générale CIB Puteaux, FR

- Analyzed **corporate actions** eligibility and executed traders' instructions, modeling event propagation in internal systems to assess P&L impact on **Swap** portfolios.
- Engineered VBA-based automation for event propagation workflows to the Front Office, ensuring high data consistency and reporting efficiency.

PROJECT

Market-Neutral Algorithmic Trading System | *Python* (*pandas, scikit-learn, statsmodels*) March 2026 – July 2026

Designed, built, and deployed a fully autonomous, regime-switching market-neutral trading system on crypto derivatives.

- Regime Detector — hourly macro filter (BTC volatility, ETH Hurst & ADX) dynamically allocating capital between the two strategies below.
- Statistical Arbitrage — cointegrated pairs trading (Engle-Granger, Rolling OLS hedge ratio) filtered by a LightGBM meta-labeling model (Lopez de Prado's Triple Barrier Method), raising win-rate from 58.7% to 88.7% out-of-sample.
- Funding Rate Carry — delta-neutral long-spot/short-perpetual positions harvesting funding payments, selected via a fee-adjusted breakeven filter.

Deployed as an autonomous cloud pipeline (Render, Supabase) with live execution and a public real-time dashboard.

Live at algo-trading-bot-d6yk.onrender.com.

EDUCATION

Master's in Quantitative Finance & Engineering Sept 2021 - July 2027
ECE Paris, FR

- Coursework: Market Pricing & Risk, VBA, **Python**, Stochastic Calculus, Interest Rate Economics, Optimal Control & Hamiltonians, **Machine Learning**, AI, **Financial Derivatives**, Graph Theory, Forecasting Techniques
- Projects: Built an XGBoost-based predictive model for equity prices, engineering technical features (RSI, MACD) and cyclical time encoding, achieving a normalized RMSE of 0.097
- Exchange Program: 5 months at University of Malaya, Kuala Lumpur, Malaysia

EXTRACURRICULAR ACTIVITIES

General Secretary Sept 2024 - Apr 2025
ECE Finance Club Paris, FR

- Organization of 15+ events, including training sessions with a Bourse Direct expert (70+ participants), a trading room simulation (80+ participants), and the Next Challenge #4 with Université Paris-Dauphine.

SKILLS & CERTIFICATIONS

Technical Skills : **Python**, VBA, SQL, Microsoft Office Suite, Bloomberg

Languages : French (native), **English** (bilingual, **TOEIC: 975**, 4 years of primary education in the USA), Spanish (school level)

Certifications : French Financial Markets Authority (AMF), Global Financing Solutions (EDHEC & SG), Financial Markets (Yale),

Machine learning in Python with scikit-learn (Inria)

Achievements : **1st place at the Finance Accelerator Investment Banking** – AmplifyME Simulation, 12th/182 at Next Challenge #4

Interests : Tennis, Running, Gym, Entrepreneurship